

Programming Using C++ Level 1

Function Parameters By Val VS By Ref

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Function Parameters

By Val

VS

By Ref

Definition

- ◆ Passing by value means that a copy of the variable is passed to the function.
- ◆ Passing by reference means that the function receives a reference to the original variable.
- ◆ The reference operator & is used with a parameter to pass it by reference.

Explanation

- ◆ In this program, the Swap() function receives A and B by reference:
void Swap(int &A, int &B)
- ◆ Because A and B are references to the original variables in main(), changing them inside the function also changes the original variables.
- ◆ The function uses a temporary variable Temp to exchange the values of A and B.

Steps in main()

1 Define the function

- ◆ Create Swap() with two reference parameters.

2 Store the first value

- ◆ Store A in the temporary variable Temp.

3 Move B into A

- ◆ Assign B to A.

4 Move the original A into B

- ◆ Assign Temp to B.

5 Call the function

- ◆ Pass A and B to Swap().

6 Display the values

- ◆ Print the values before and after the swap.

💡 Example

- ◆ Function using pass by reference:

```
void Swap(int &A, int &B)
{
    int Temp;
    Temp = A;
    A = B;
    B = Temp;
}
```

- ◆ If:

```
A = 10
B = 20
```

- ◆ After calling:

```
Swap(A, B);
```

- ◆ The original variables become:

```
A = 20
B = 10
```

➡ Result

- ◆ If the user enters: A = 10 B = 20

- ◆ The program displays:

Before Swap A= 10, B= 20

After Swap inside function A= 20, B= 10

Note

- ◆ & in a function parameter means that the parameter is a reference to the original variable.
- ◆ Pass by reference is useful when a function needs to modify the original variables.
- ◆ The Swap() function must use references here; if A and B were passed by value, the original values in main() would not change.

⊘ Warning

◆ Passing by reference allows the function to modify the original variable, so use it when that behavior is intended.



Comparison

▨ Pass By Value

- ✓ A copy of the variable is passed to the function.
- ✓ Changes inside the function do not affect the original variable.
- ✓ Uses a separate value for the parameter.

▤ Pass By Reference

- ✓ The function works with the original variable.
- ✓ Changes inside the function affect the original variable.
- ✓ Uses & in the parameter declaration.

✦✦ Summary

- ✓ Pass by value sends a copy of the variable to the function.
- ✓ Pass by reference allows the function to work with the original variable.
- ✓ The & operator is used to pass parameters by reference.

Thank you 😊